

團體程式作業

分工表：

內容	學號	姓名
Chapter7	112820003	辛政隆
Chapter8 Safety algorithm 實 作	112820025	陳翊揚
Chapter8 (Customer thread)建 立與迴圈	112820018	許景喬
文件彙整	112820042	杜詠霖

CH7 Project 4 : The Producer-Consumer Problem

建議使用 Linux、WSL、macOS 或支援 GCC / Make 的環境

編譯方式：在 Maefile 所在位置 make

執行方式：make 編譯組

./producer_consumer 執行程式

執行結果：如 Ch7/output.txt 所示

CH8 Programming Project : Banker's Algorithm

建議使用 Linux、WSL、macOS 或支援 GCC / Make 的環境

編譯方式：在 Maefile 所在位置 make

執行方式：make 編譯組

./banker 執行程式 + 參數或預設值

預設值：demo、demo_request、demo_sim

命令列參數：

```

banker -n <processes> -m <resources>
    -avail <r0> <r1> ...
    -alloc <p0r0> <p0r1> ... <p1r0> <p1r1> ...
    -max   <p0r0> <p0r1> ... <p1r0> <p1r1> ...
    [-request <pid> <r0> <r1> ...]
    [-sim]

```

-n	Process 數量
-m	Resource type 數量
-avail	Available vector，需要輸入 m 個值
-alloc	Allocation matrix，需要輸入 n * m 個值
-max	Maximum demand matrix，需要輸入 n * m 個值
-request	指定 process 的資源請求
-sim	啟用多執行緒模擬模式
-h / --help	顯示使用說明

執行結果：

demo

```

root@LAPTOP-E3J5P8BM:/mnt/e/大三/作業系統/HW1/114-2_Operator_System_Team_Project/HW3/Ch8# ./banker demo
[No arguments given - loading textbook default example]

=== Banker's Algorithm ===

Processes: 5    Resource Types: 3

Allocation:
    R0  R1  R2
P0   0   1   0
P1   2   0   0
P2   3   0   2
P3   2   1   1
P4   0   0   2

Max:
    R0  R1  R2
P0   7   5   3
P1   3   2   2
P2   9   0   2
P3   2   2   2
P4   4   3   3

Need:
    R0  R1  R2
P0   7   4   3
P1   1   2   2
P2   6   0   0
P3   0   1   1
P4   4   3   1

Available:
    R0  R1  R2
    3   3   2

--- Initial Safety Check ---
System is in a SAFE state.
Safe sequence: P1 -> P3 -> P4 -> P0 -> P2

```

demo_request

```

root@LAPTOP-E3J5P8BM:/mnt/e/大三/作業系統/HW1/114-2_Operator_System_Team_Project/HW3/Ch8# ./banker demo_request
[No arguments given - loading textbook default example]

=== Banker's Algorithm ===

Processes: 5    Resource Types: 3

Allocation:
    R0  R1  R2
P0   0   1   0
P1   2   0   0
P2   3   0   2
P3   2   1   1
P4   0   0   2

Max:
    R0  R1  R2
P0   7   5   3
P1   3   2   2
P2   9   0   2
P3   2   2   2
P4   4   3   3

Need:
    R0  R1  R2
P0   7   4   3
P1   1   2   2
P2   6   0   0
P3   0   1   1
P4   4   3   1

Available:
    R0  R1  R2
    3   3   2

--- Initial Safety Check ---
System is in a SAFE state.
Safe sequence: P1 -> P3 -> P4 -> P0 -> P2

```

demo_sim

```
root@LAPTOP-E3J5P8BM:/mnt/e/大三/作業系統/HW1/114-2_Operator_System_Team_Project/HW3/Ch8# make demo_sim
./banker \
    -n 5 -m 3 \
    -avail 3 3 2 \
    -alloc 0 1 0 2 0 0 3 0 2 2 1 1 0 0 2 \
    -max 7 5 3 3 2 2 9 0 2 2 2 2 4 3 3 \
    -sim
=== Banker's Algorithm ===

Processes: 5   Resource Types: 3

Allocation:
      R0  R1  R2
P0    0   1   0
P1    2   0   0
P2    3   0   2
P3    2   1   1
P4    0   0   2

Max:
      R0  R1  R2
P0    7   5   3
P1    3   2   2
P2    9   0   2
P3    2   2   2
P4    4   3   3

Need:
      R0  R1  R2
P0    7   4   3
P1    1   2   2
P2    6   0   0
P3    0   1   1
P4    4   3   1

Available:
      R0  R1  R2
      3   3   2

--- Initial Safety Check ---
System is in a SAFE state.
```

```
--- Starting Multithreaded Simulation ---
Press Ctrl+C to stop.

>> P0 requesting resources...
<< P0: Request DENIED/WAIT (Insufficient or Unsafe).
>> P1 requesting resources...
<< P1: Request GRANTED.
>> P4 requesting resources...
<< P4: Request DENIED/WAIT (Insufficient or Unsafe).
>> P2 requesting resources...
<< P2: Request DENIED/WAIT (Insufficient or Unsafe).
>> P2 releasing resources...
<< P2: Resources released.
>> P4 releasing resources...
<< P4: Resources released.
>> P1 releasing resources...
<< P1: Resources released.
>> P1 requesting resources...
<< P1: Request GRANTED.
>> P2 requesting resources...
<< P2: Request DENIED/WAIT (Insufficient or Unsafe).
>> P0 requesting resources...
<< P0: Request DENIED/WAIT (Insufficient or Unsafe).
^Cmake: *** [Makefile:33: demo_sim] Interrupt
```